UNIFIED MECHANICS

THE REALISATION COMPILER

From Relational Machine to Physical Prediction

FORMAL APPLICATION PROTOCOL AND FIVE-STAR CARBON-OXYGEN WORKED EXAMPLE

The physical machine must be reconstructed before the mathematics is allowed to name its output.

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16 July 2026

APPLICATION DOCUMENT | FROZEN WORKED EXAMPLE

THE GOVERNING SENTENCE

APPLICATION RULE

Do not ask which Unified Mechanics number resembles the observed result. Reconstruct the Whole, Parts, Boundary, Local Conditions, Process, Clocks and Records; derive the ordinary physical machine; compress its update law; and only then ask whether that machine realises an already established UM operation.

Abstract

This document formalises a repeatable method for applying Unified Mechanics to physical systems without beginning from numerical resemblance. A realised machine is defined as a network of systems whose outputs alter one another's local conditions, boundaries or admissible processes. The method therefore begins with the seven-part realised-system schema, recovers the conserved currency and coupling graph, derives the ordinary physical equations, and only then compresses the machine into dimensionless coordinates. A UM bridge is admitted only when the compressed physical update law is equivalent to an already established UM operation under a stated condition. The leading example is helium-burning stellar carbon-oxygen production. The example derives a carbon-lineage processing machine, compresses it to a retention coordinate x, continued-processing pressure chi and Boundary selectivity beta, and identifies the exact condition required for the stellar machine to instantiate the UM recurrence. A five-star terminal record screen is included to demonstrate the governance rule: a numerical average may not rescue an incorrect choice of Whole, and a terminal abundance match cannot substitute for a machine-level test.

Document status

STATUS

The application protocol is a formal methodological proposal built from the governing UM system schema. The algebraic compression of the stellar process equations is exact under the stated definitions. The stellar-to-UM bridge is conditional and remains to be tested on time-resolved stellar evolution tracks. The five-star table is a terminal-record screening example, not a completed validation of the bridge.

Claim grammar

Grade	Meaning in this document
DEFINED	Notation or role fixed by declaration.
DERIVED	Follows algebraically from the declared physical process equations.
CONDITIONAL	Follows if an explicit physical bridge condition is satisfied.
PROPOSED	A physical realisation or execution method offered for testing.
RECORD	A published or model-mediated observational input.
OPEN	Required calculation or dataset not yet executed.

CONTENTS

PART I

THE RELATIONAL MACHINE

PLAIN-LANGUAGE RULE

A system does not define itself by speaking. Its present relations restrict what it can do next, and the state that follows preserves a physical record of those restrictions.

1. The puddle is a record

Imagine rain falling into a depression in the ground. A drop may wonder why the puddle fits the ground so perfectly, as though the depression had been designed for this exact water. The fit is not mysterious. Water, terrain, gravity, surface tension, absorption, temperature and time act together until only a locally admissible shape remains.

The puddle is therefore not merely an object sitting in a container. It is the terminal state of a coupled process. Its shoreline, depth and wet boundary are records of the relations that produced it.

$$\text{puddle shape} = \text{rain} + \text{terrain} + \text{gravity} + \text{material response} + \text{time} + \text{boundary exchange} \tag{1.1}$$

THE IMPORTANT DISTINCTION

The hole does not predict the exact drop, and the drop does not independently choose the hole. The realised shape is selected by the coupled machine.

2. The Relational Machine Principle

At effective scale ell, let each realised component system be represented by the seven-part schema

$$S\_i = (X\_i, P\_i, B\_i, L\_i, \text{Gamma}\_i, \text{tau}\_i, R\_i) \tag{2.1}$$

where X is the retained Whole, P the relevant Parts, B the Actual Boundary, L the Local Conditions, Gamma the admissible Process family, tau the relevant Clocks and R the preserved Records. This is the governing realised-system structure of the Unified Mechanics corpus.

A machine requires more than a list of component systems. The output or record of one component must become a condition, boundary input or process constraint for another. Define a coupling map

$$C\_ij : R\_i \text{ -> } L\_j \text{ union } B\_j \text{ union Gamma}\_j. \tag{2.2}$$

The realised machine is then the network

$$M = (\{S\_1, ..., S\_n\}, \{C\_ij\}). \tag{2.3}$$

RELATIONAL MACHINE PRINCIPLE

A realised machine is a collection of systems whose outputs alter one another's local conditions, boundaries or admissible processes, while the coupled whole retains an identifiable organisation through time and leaves reconstructable records.

The machine is not equivalent to the arithmetic sum of its objects. Its coupling pattern is part of the physical object being described.

$$M \text{ != sum\_i } S\_i\text{; the coupling relations are constitutive parts of } M. \tag{2.4}$$

3. Operational self-description without mysticism

The phrase "a system describes itself" is operational, not psychological. A system describes itself when its retained state and local conditions select transformations that leave reconstructable differences in later states or external records.

$$S_{(n+1)} = F(S_n, E_n), \qquad R_n = M(S_n), \qquad E_{(n+1)} = E_n \text{ plus } C(R_n). \tag{3.1}$$

The later state contains information about the earlier relations because those relations physically constrained the transition. A compatible observer can work backward from the record, but the observer did not manufacture the history.

NON-MYSTICAL TRANSLATION

"Everything defines itself through its relationships" means: the relations among a system's parts determine its admissible transformations, and the resulting transformations preserve records from which those relations can be inferred.

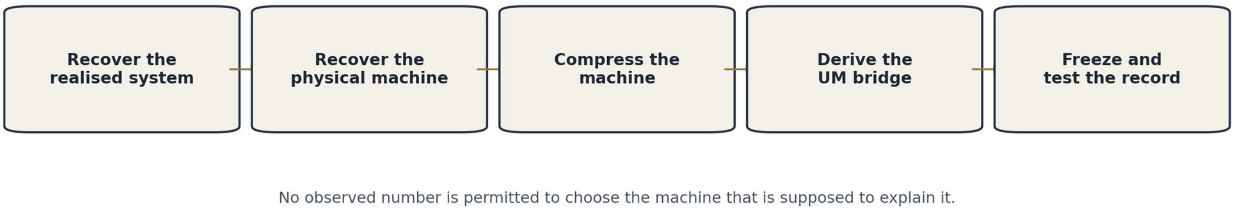

Figure 1. The UM application order. A familiar measured value appears only at the final stage.

PART II

THE UM REALISATION COMPILER

4. The four passes

PASS I - RECOVER THE SYSTEM
Declare the Whole, Parts, Boundary, Local Conditions, Process, Clocks and Records without mentioning the desired UM number.

PASS II - RECOVER THE MACHINE
Identify the conserved or inherited currency, map the couplings among subsystems and write the ordinary physical state equations.

PASS III - COMPRESS AND BRIDGE
Derive the minimum dimensionless coordinate that preserves the relevant distinction, then compare its physical update law with an already established UM operation.

PASS IV - FREEZE AND TEST
Freeze the bridge, observable, data selection, residual definition and rejection rule before inspecting the target records.

5. The fourteen execution steps

1. Question freeze

State the physical question without proposing a target number.

2. Whole

Choose the smallest retained system that contains the complete causal process.

3. Parts

List only the components whose relations realise the chosen Whole.

4. Boundary

Identify the scale-, channel- and clock-dependent cut where retention or transfer is resolved.

5. Local Conditions

Specify the variables that select the actual behaviour here and now.

6. Process

Write the transformations and transport laws that physically occur.

7. Clocks

Separate formation, processing, propagation, relaxation and record times.

8. Records

State what is directly preserved and what is inferred through a model.

9. Currency

Identify what is conserved, inherited, exchanged or counted through the process.

10. Coupling graph

Show which output becomes which later input.

11. Physical equations

Derive the standard state-update equations before using UM notation.

12. Compression

Construct dimensionless coordinates from the physical equations, not from observed resemblance.

13. Bridge

Derive the condition under which the compressed physical map equals an established UM map.

14. Freeze and test

Generate the observable, freeze the sample and apply the fixed rejection rule.

6. Numerical governance and the fixed residual ceiling

The residual ceiling is not used to make an unfinished mapping appear successful. It applies only after the complete physical pipeline and the UM bridge have been recovered.

$$\text{epsilon\_*} = r^3 = 0.029508497\ldots \tag{6.1}$$

$$\text{delta} = (O - P) / P. \tag{6.2}$$

A completed numerical prediction is classified by the following frozen rule:

- abs(delta) <= r^3: resolved prediction within the universal record envelope.
- abs(delta) > r^3: the physical mapping, chosen Whole or recovered pipeline has failed or remains incomplete.
- observational uncertainty > r^3: the test is unresolved; it is not counted as a pass.
- a later correction is admissible only when independently forced by the seven-part machine, never because it improves the number.

RECOVERY RULE

An out-of-band result sends the analysis backward through the pipeline. It does not permit a wider tolerance. If the complete independently required machine remains outside the fixed envelope, that physical realisation is rejected.

PART III

LEADING WORKED EXAMPLE: THE FIVE-STAR CARBON-OXYGEN MACHINE

PURPOSE OF THE EXAMPLE

The example is not designed to make a known abundance resemble a golden number. It demonstrates how a familiar physical output is converted into a machine-level question, an exact compression and a falsifiable bridge.

7. Freeze the question

The incorrect question is: "Does the carbon-oxygen ratio of a star look like r?" That question begins with the output and searches backward for a label.

The frozen machine question is:

STELLAR PROCESSING QUESTION

For a declared helium-burning stellar system, what physical machine creates carbon, processes some carbon onward into oxygen, transports the products, regulates the reaction environment and preserves a later carbon-oxygen record? Does the compressed update law of that complete machine instantiate a pre-existing UM recurrence?

8. Recover the seven-part stellar system

Role	Stellar realisation	Why it matters
Whole	The convectively connected helium-burning core during a declared processing interval.	The whole star contains later shells, envelopes and records that are not part of the same immediate reaction machine.
Parts	Helium fuel; carbon; oxygen; plasma; released energy; convective/radiative transport; surrounding layers.	Carbon and oxygen are nested material states inside one coupled stellar machine.
Boundary	The time-dependent edge of the connected burning/mixing region.	It determines what remains in the core and what is exchanged with surrounding layers.
Local Conditions	Temperature, density, helium abundance, pressure, composition, opacity, gradients and mixing state.	These select the reaction and transport rates.
Process	Triple-alpha carbon creation; carbon-alpha oxygen conversion; energy release; mixing; expansion and transport.	The final ratio is produced by a feedback network, not a single binary choice.
Clocks	Fuel depletion, nuclear processing, convective turnover, thermal relaxation, progenitor evolution and white-dwarf cooling.	Different carbon cohorts remain exposed to continued processing for different durations.
Records	Asteroseismic periods, inferred internal chemical profiles, ejecta spectra and evolutionary model outputs.	The central composition is model-mediated rather than directly counted.

8.1 The inherited currency

The process itself determines the currency. One triple-alpha event creates one carbon lineage. A later carbon-alpha event moves that same lineage from the carbon class into the oxygen class. The lineage count is therefore

$N(t) = C(t) + O(t).$	(8.1)

This is not chosen because it produces a convenient ratio. It is the identity preserved by the stated reaction network. Mass weighting is applied later when converting the lineage state into a mass-fraction record.

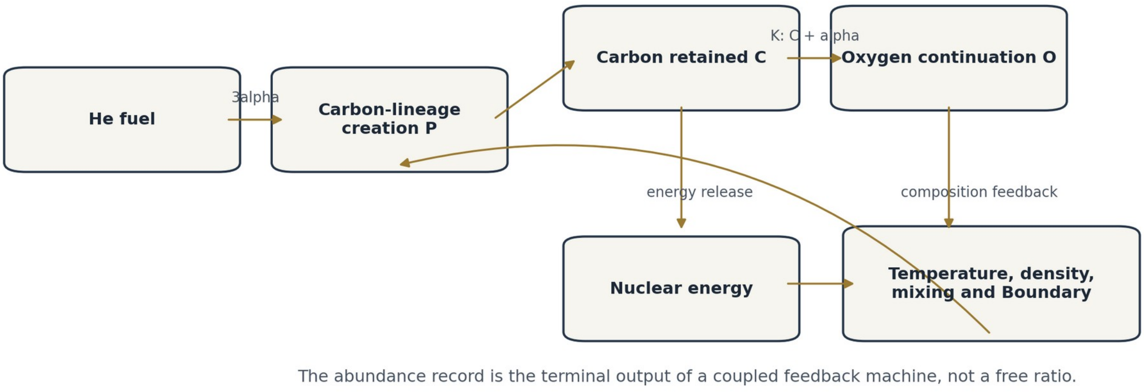

Figure 2. The stellar abundance is the terminal record of a coupled chemical, energetic and boundary-regulation machine.

9. Derive the ordinary physical machine

The two principal lineage-changing reactions are

3 alpha -> carbon-12 + gamma,	(9.1)
carbon-12 + alpha -> oxygen-16 + gamma.	(9.2)

Let P(t) be the integrated carbon-lineage production rate, K(t) the integrated conversion rate from carbon to oxygen, and Phi_C and Phi_O the carbon and oxygen fluxes through the selected Boundary. The ordinary bookkeeping is

dC/dt = P - K - Phi_C,	(9.3)
dO/dt = K - Phi_O,	(9.4)
dN/dt = P - Phi_C - Phi_O.	(9.5)

The rates P and K are functions of the local stellar state. The reaction output changes the energy budget; energy changes temperature, pressure, expansion and mixing; those conditions feed back into the reaction rates and Boundary. Schematically,

composition -> reaction -> energy -> local conditions -> Boundary/mixing -> composition.	(9.6)

MACHINE-LEVEL CONSEQUENCE

The final carbon-oxygen record is not generated by one instantaneous split. It is an integrated survival and continuation history inside a self-regulating stellar system.

10. Compress the machine

Define the retained carbon-lineage coordinate

$$x = C / (C + O) = C / N. \tag{10.1}$$

Differentiate x using equations (9.3)-(9.5). The result is

$$dx/dt = [P(1 - x) - K - (1 - x)Phi\_C + x\ Phi\_O] / N. \tag{10.2}$$

Now define a production-normalised processing clock, a continued-processing pressure and a Boundary-selectivity coordinate:

$$d\ tau = (P/N)\ dt, \tag{10.3}$$

$$chi = K / (P\ x), \tag{10.4}$$

$$beta = [(1 - x)Phi\_C - x\ Phi\_O] / P. \tag{10.5}$$

Substitution gives the exact compressed machine

$$dx/d\ tau = 1 - (1 + chi)x - beta. \tag{10.6}$$

DERIVED RESULT

The complete stellar machine has been reduced to three physically typed coordinates: x is retained lineage, chi is continued-processing pressure, and beta is composition-selective Boundary transport. No UM number has yet been assigned.

10.1 Neutral Boundary condition

If the selected Boundary transports carbon and oxygen proportionally to their current populations,

$$Phi\_C / C = Phi\_O / O, \tag{10.7}$$

then beta = 0 and the machine becomes

$$dx/d\ tau = 1 - (1 + chi)x. \tag{10.8}$$

For approximately fixed chi, the instantaneous equilibrium coordinate is

$$x\_eq = 1 / (1 + chi). \tag{10.9}$$

11. Derive the candidate UM bridge

The internal UM recurrence is taken as already established within the mathematical corpus:

$$u\_(n+1) = 1 / (2 + 4u\_n). \tag{11.1}$$

The stellar coordinate x may not be identified with u because both are verbally described as retention. The physical equilibrium map must equal the UM map. From (10.9) and (11.1),

$$1 / (1 + chi\_n) = 1 / (2 + 4x\_n). \tag{11.2}$$

Therefore the exact bridge condition is

$$chi\_n = 1 + 4x\_n, \tag{11.3}$$

$$K\_n / (P\_n\, x\_n) = 1 + 4x\_n. \tag{11.4}$$

STELLAR REALISATION CRITERION

A helium-burning stellar machine instantiates the selected UM recurrence only if its carbon-to-oxygen continuation pressure, measured relative to carbon-lineage production and current carbon retention, satisfies K/(Px) = 1 + 4x through the declared closed processing regime.

This condition was not selected by examining a final C/O abundance. It is forced by requiring equivalence between the compressed physical map and the pre-existing UM map.

12. Freeze the numerical consequences

Under beta = 0 and chi = 1 + 4x, the continuous compressed process is

$$dx/d\,tau = 1 - 2x - 4x^2. \tag{12.1}$$

The stable positive fixed point satisfies

$$4x^2 + 2x - 1 = 0. \tag{12.2}$$

$$x\_* = r = (sqrt(5) - 1)/4 = 0.309016994375. \tag{12.3}$$

Because x is a lineage-number fraction, the terminal nucleus-count prediction is

$$O/C = (1 - r)/r = sqrt(5) = 2.236067977.... \tag{12.4}$$

The corresponding oxygen-to-carbon mass ratio is

$$M\_O/M\_C = 16O/(12C) = 4sqrt(5)/3 = 2.981423970.... \tag{12.5}$$

$$X\_C = 0.251166419, \quad X\_O = 0.748833581 \quad \text{(normalised C/O ash mass fractions).} \tag{12.6}$$

These numerical consequences are conditional outputs of the bridge. They are not accepted as stellar predictions unless the machine criterion itself is satisfied.

13. Five-star terminal record screen

The first screen uses five published, model-mediated white-dwarf core records. Four are Kepler ZZ Ceti stars analysed with fully evolutionary carbon-oxygen core models; CBS 114 is an independent DBV asteroseismic case. The table tests only the terminal shortcut x_f = r. It cannot test the time-resolved bridge because the published record does not provide P(t), K(t), Phi_C(t) and Phi_O(t).

PROVISIONAL INPUT WARNING

The four Kepler x values are the working central C/O coordinates previously transcribed from the corresponding best-fit evolutionary model profiles. Before journal publication, they should be replaced by direct machine-readable profile extraction from the model files or an author-supplied table. CBS 114 follows directly from the published central oxygen fraction X_O = 0.61 under the two-component C/O core approximation.

Star	x_star	O/C	Signed residual	3% result	Record source
KIC 11911480	0.29382	2.403	-4.92%	OUTSIDE	Romero et al. (2017) model profile
SDSS J113655.17+040952.6	0.30191	2.312	-2.30%	INSIDE	Romero et al. (2017) model profile

KIC 4552982	0.33435	1.991	+8.20%	OUTSIDE	Romero et al. (2017) model profile
GD 1212	0.23660	3.227	-23.43%	OUTSIDE	Romero et al. (2017) model profile
CBS 114	0.39000	1.564	+26.21%	OUTSIDE	Metcalfe & Handler (2002)

Five-star mean: x_bar = 0.311336; residual from r = +0.750%.

14. Interpretation of the screen

The per-star terminal claim x_star = r fails for four of the five provisional records under the fixed r^3 ceiling. The mean happens to lie within one percent of r, but the mean may not rescue the per-star claim. If the Whole is one star, each star must be evaluated as one machine.

The ensemble mean can become a legitimate object only if the ensemble is declared as the Whole before record inspection, its membership and weights are frozen, and a population-level coupling process is derived that predicts centring on r. None of those conditions is supplied by taking an average after seeing the five values.

WHAT THE FAILED SHORTCUT TEACHES

The record screen does not reject the application protocol. It rejects the unsupported shortcut from final central abundance directly to x = r. The correct test is the bridge trajectory K/(Px) = 1 + 4x, with Boundary selectivity and processing clocks explicitly recovered.

14.1 Record status

- The observed pulsation periods are physical records.
- The internal C/O profiles are established-model inferences from those records, not direct nucleus counts.
- The coordinate x is a derived compression of the inferred profile.
- The equality x = r is a conditional UM consequence only after the bridge is established.
- The present five-star table is therefore a terminal diagnostic, not a machine validation.

15. Full five-track machine execution plan

A decisive test requires time-resolved progenitor evolution tracks rather than only white-dwarf terminal profiles. The five tracks should be selected to produce remnants spanning the observed mass range of the worked sample. The same frozen input physics must be used for all tracks.

15.1 Frozen inputs

- One stellar evolution code and one version for all runs.
- One declared metallicity grid or one fixed metallicity; no target-driven switching.
- One nuclear reaction-rate library, including fixed triple-alpha and carbon-alpha rates.
- One convection and overshoot prescription.
- One spatial and temporal resolution policy.
- Progenitor masses selected to span remnant mass, not to improve C/O agreement.
- No parameter may be changed after inspecting the UM bridge residual.

15.2 Required time-series records

- stellar age and model number
- central and profile temperature and density
- helium-4, carbon-12 and oxygen-16 abundances
- convective core mass and Boundary location
- integrated triple-alpha lineage-production rate P(t)
- integrated carbon-alpha continuation rate K(t)

- carbon and oxygen fluxes across the chosen Boundary
- nuclear luminosity and total luminosity
- mixing timescale and helium-depletion clock
- final white-dwarf C/O profile and pulsation-ready model

15.3 Derived machine ledger

x(t) = C/(C + O),	(15.1)
chi(t) = K/(P x),	(15.2)
beta(t) = [(1 - x)Phi_C - x Phi_O]/P,	(15.3)
Delta_bridge(t) = [chi(t) - (1 + 4x(t))] / [1 + 4x(t)].	(15.4)
Delta_state(t) = [dx/dtau - (1 - 2x - 4x^2)] / max(1, abs(dx/dtau)).	(15.5)

15.4 Pass conditions

- Machine bridge: |Delta_bridge| <= r^3 across the independently identified closed helium-processing interval.
- Boundary: beta is either within the fixed envelope of zero or follows a separately derived UM Boundary law.
- Dynamical state: the measured compressed derivative follows 1 - 2x - 4x^2 within the declared numerical and record resolution.
- Terminal record: the predicted post-processing profile maps through the white-dwarf evolution and asteroseismic record model without an out-of-band residual.
- Any uncertainty larger than r^3 makes that component unresolved, not successful.

15.5 Failure diagnosis

Observed failure	Pipeline diagnosis
beta is large	The chosen Boundary is selective, moving, or defined at the wrong scale.
chi has the right shape but wrong amplitude	The physical process may be incomplete, or the selected currency/clock may be incorrectly normalised.
bridge works internally but terminal C/O fails	Later processing, transport, white-dwarf evolution or the record map is missing.
all complete pipeline elements remain outside r^3	Reject this stellar realisation of the UM recurrence.
record uncertainty exceeds r^3	The dataset cannot resolve the proposed law.

PART IV

GENERAL APPLICATION STANDARD

16. Reusable execution worksheet

A. QUESTION FREEZE What process is being explained? What output is forbidden from influencing the machine definition?
B. WHOLE What is the smallest retained system containing the complete causal sequence?
C. PARTS Which components and roles realise that Whole?
D. BOUNDARY Where, for the selected scale, channel and clock, is retention or transfer resolved?
E. LOCAL CONDITIONS Which variables select the actual process?
F. PROCESS What transformations, fluxes and feedbacks occur?
G. CLOCKS Which formation, processing, propagation, relaxation and observation times are distinct?
H. RECORDS What is directly stored, and what is inferred?
I. CURRENCY What identity, mass, count, action, energy or record distinction is inherited?
J. COUPLING GRAPH Which output modifies which later condition, Boundary or process?
K. PHYSICAL EQUATIONS What ordinary equations express the machine without UM?
L. COMPRESSION Which dimensionless coordinates preserve the relevant process distinction?
M. UM OPERATION Which already-proved UM map has the same operational role?
N. BRIDGE CONDITION What equation must hold for the physical map and UM map to be equivalent?
O. OBSERVABLE How does the bridge produce a record through a declared channel and inference model?
P. FREEZE Which sample, units, weighting, uncertainty and rejection rules are fixed in advance?

17. Bridge proof requirements

A bridge is not a metaphorical dictionary. It is a demonstration that two update laws perform the same operation under declared hypotheses. Every physical bridge must state:

- Domain: the physical states for which the mapping is defined.
- Codomain: the UM coordinate or operation being realised.
- Type preservation: quantities compared must represent the same physical currency and transformation role.
- Update equivalence: the physical evolution map must reduce to the UM map, not merely share a fixed point.
- Boundary and clock conditions: the scale and interval over which equivalence is claimed.
- Record map: the observable generated by the physical state and the inference used to recover it.
- Kill condition: a result that rejects the bridge rather than being absorbed as unmodelled complexity.

> **NO FIXED-POINT SHORTCUT**
> Two unrelated machines can share a numerical fixed point. A valid application must recover the update law that produces the fixed point.

18. Falsification and recovery rules

1. Freeze the current physical mapping and calculate its signed residual.
2. If the residual lies outside r^3, do not widen the band.
3. Audit Whole, Parts, Boundary, Local Conditions, Process, Clocks and Records in that order.
4. Admit a missing component only when the physical system independently requires it.
5. Re-derive the physical machine and bridge from the beginning; do not patch the terminal formula.
6. Record both the failed and revised mappings in the provenance ledger.
7. If a complete independently required pipeline remains outside r^3, reject the realisation.

18.1 Forbidden recovery moves

- Changing the Whole after seeing that an ensemble average fits.
- Switching from number fraction to mass fraction because one is closer.
- Assigning Light, Boundary or Matter labels from numerical resemblance.
- Adding a correction with no role in the seven-part machine.
- Calling a large observational uncertainty agreement.
- Using the r^3 envelope before the physical bridge has been completed.

19. Final application theorem

> **UM APPLICATION THEOREM - METHODOLOGICAL FORM**
> A numerical relation is an admissible Unified Mechanics physical prediction only when: (i) a nonempty realised system is recovered through Whole, Parts, Boundary, Local Conditions, Process, Clocks and Records; (ii) the machine's inherited currency, couplings and ordinary update law are derived independently of the target number; (iii) a dimensionless compression preserves the relevant operation; (iv) an explicit bridge proves equivalence to an established UM map under declared conditions; and (v) the resulting observable is frozen and tested through a stated record channel. Numerical resemblance without this chain is not an application of Unified Mechanics.

The application method can be remembered in one sentence:

$$\text{Recover the machine -> compress its action -> prove the bridge -> freeze the record.} \tag{19.1}$$

The puddle is not important because it resembles a number. It is important because its shape is a recoverable record of rain, ground, gravity, boundary response and time. The same rule now governs every UM application.

APPENDICES

Appendix A. Exact stellar derivation

Starting from $x = C/N$ with $N = C + O$,

$$dx/dt = (N\, dC/dt - C\, dN/dt) / N^2. \tag{A.1}$$

Substitute $dC/dt = P - K - \Phi_C$ and $dN/dt = P - \Phi_C - \Phi_O$:

$$dx/dt = \{N(P - K - \Phi_C) - C(P - \Phi_C - \Phi_O)\} / N^2. \tag{A.2}$$

Using $C = xN$ and collecting terms gives

$$dx/dt = [P(1 - x) - K - (1 - x)\Phi_C + x\Phi_O] / N. \tag{A.3}$$

Divide by $d\tau/dt = P/N$ and introduce χ and β:

$$dx/d\tau = 1 - x - K/P - [(1 - x)\Phi_C - x\Phi_O]/P, \tag{A.4}$$

$$dx/d\tau = 1 - (1 + \chi)x - \beta. \tag{A.5}$$

The bridge $\chi = 1 + 4x$ and $\beta = 0$ produces

$$dx/d\tau = 1 - 2x - 4x^2. \tag{A.6}$$

The derivative of the right-hand side at the positive fixed point is negative, so the continuous fixed point is locally attracting:

$$\frac{d}{dx}(1 - 2x - 4x^2)\Big|_{x=r} = -2 - 8r = -2\sqrt{5} < 0. \tag{A.7}$$

Appendix B. Five-star calculation ledger

For each working record x_{star}, the terminal diagnostics are

$$O/C = (1 - x_{star})/x_{star}, \quad \delta_x = (x_{star} - r)/r. \tag{B.1}$$

Star	x_star	C/O	O/C	delta_x
KIC 11911480	0.29382	0.4161	2.4034	-0.049179
SDSS J113655.17+040952.6	0.30191	0.4325	2.3122	-0.022999
KIC 4552982	0.33435	0.5023	1.9909	+0.081979
GD 1212	0.23660	0.3099	3.2265	-0.234346
CBS 114	0.39000	0.6393	1.5641	+0.262067

The values above are retained exactly as the frozen working ledger for this application document. They are not presented as a replacement for direct profile-file extraction.

Appendix C. Stellar evolution instrumentation specification

A practical execution can be implemented in MESA or an equivalent stellar evolution code. The model must expose both global history columns and radial profile data during core-helium burning.

C.1 Suggested global history fields

- model number and star age
- star mass and helium-core mass
- central temperature and density
- central he4, c12 and o16 mass fractions
- convective core mass
- nuclear luminosity, total luminosity and neutrino losses
- time step and helium-depletion fraction
- custom integrated P, K, Phi_C and Phi_O columns

C.2 Suggested profile fields

- mass coordinate and radius
- temperature, density and pressure
- he4, c12 and o16 profiles
- mixing type and diffusion coefficient
- energy generation and local reaction rates
- convective velocity and Boundary location
- composition fluxes where available

C.3 Processing interval

The closed processing interval must be selected by a physical criterion, such as the onset of a connected helium-burning core and a frozen helium-depletion endpoint. It may not be selected by searching for the interval that minimises the UM residual.

Appendix D. Source and provenance ledger

1. Shields, Joseph. Monograph I - A World of Distinctions (Main Edition). Unified Mechanics, 15 July 2026.

2. Shields, Joseph. Monograph II - Universopedia: The Observational Standard. Unified Mechanics, 14 July 2026.

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5. Metcalfe, T. S., and G. Handler. "The carbon-12(alpha,gamma)oxygen-16 Nuclear Reaction Rate from Asteroseismology of the DBV White Dwarf CBS 114." arXiv:astro-ph/0208225 (2002).

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8. Handler, G., T. S. Metcalfe and M. A. Wood. "The asteroseismological potential of the pulsating DB white dwarf stars CBS 114 and PG 1456+103." arXiv:astro-ph/0205459 (2002).

END OF APPLICATION PROTOCOL